5 **(a)** The velocity v may be resolved into two components – one parallel to the magnetic field B, which is $v\cos\theta$, and the other perpendicular to B, which is $v\sin\theta$. **B0**

$v\sin\theta$ results in a circular motion of the proton in the plane perpendicular to B. **B1**

Using Fleming's Left-hand Rule, the centripetal force is directed into the page initially. **B1**

$v\cos\theta$ causes the proton / circle to move at constant speed in the same direction as B. **B1**

Consequently, the proton moves in a helical path. **B1**

(b) **(i)** **A1**

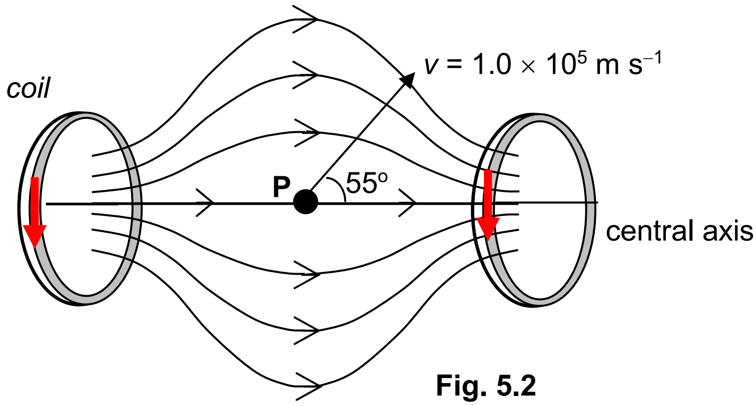


Fig. 5.2

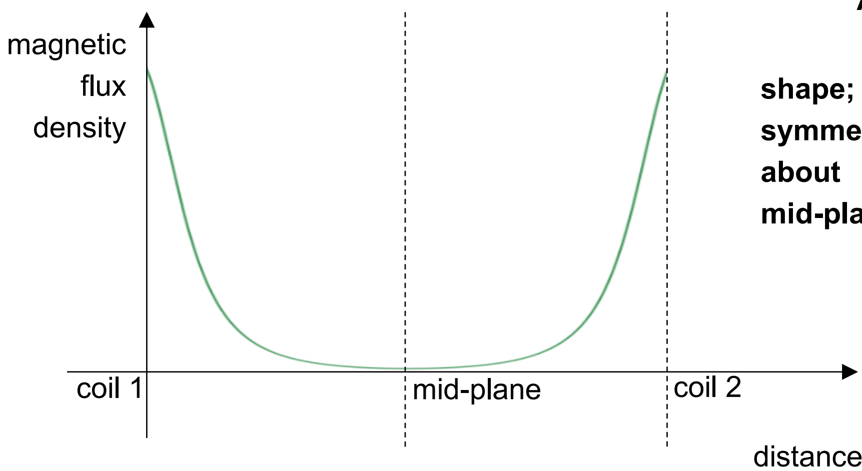
(ii) Using $F = Bqv\sin\theta$, we get

C1

$$F = 3.8\times10^{-4}\left(1.6\times10^{-19}\right)\left(1\times10^{5}\right)\sin 55^{0} = 4.98\times10^{-18}\text{ N}$$ **A1**

$$= 5.0\times10^{-18}\text{ N} \quad (\textit{to 2 sf})$$

(iii) **A1**



shape; symmetry about mid-plane

6 **(a)** **(i)** The direction of induced e.m.f. produces effects that oppose the change